The obvious next question is: What are the possible outcomes of measuring **X**, and what are the states in which it has a definite (predictable) value? In other words, what are its eigenvalues and eigenvectors? We'll start with **X**. The eigen-equation for **X** is

$$\mathbf{X}|\Psi\rangle = x_0|\Psi\rangle,$$

where the eigenvalue is denoted by x_0. In terms of wave functions, this becomes

$$x\psi(x) = x_0\psi(x). \qquad (8.11)$$

This last equation seems strange. How can x times a function be proportional to the same function? On the face of it, this seems impossible. But let's pursue it. We can rewrite Eq. 8.11 in the form

$$(x - x_0)\psi(x) = 0.$$

Of course, if a product is zero, then at least one of the factors must be zero. But the other factors may be different from zero. Thus, if $x \neq x_0$, then $\psi(x) = 0$. That's a very strong condition. It says that for a given eigenvalue x_0, the function $\psi(x)$ can be nonzero at only one point, namely at

$$x = x_0.$$

For an ordinary continuous function this condition would be deadly: no sensible function can be zero everywhere except at one point, and be nonzero only at that point. But that is exactly the property of the Dirac delta function

$$\delta(x - x_0).$$